Lecture Notes in Physics 821

An Introduction to the Confinement Problem

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Chapter 2
Global Symmetry, Local Symmetry, and the Lattice

Confinement in non-abelian gauge theory involves the idea that the vacuum state is disordered at large scales; our best evidence that this is true comes from Monte Carlo simulations of lattice gauge theories. So to begin with, I need to explain what is meant by

- a disordered state,
- a lattice gauge theory,
- a Monte Carlo simulation.

I also need to explain, since this is a book about the confinement problem, what is meant by the word "confinement." That, however, is a surprisingly subtle, and even controversial issue, which will be deferred to the next chapter.

2.1 Global Symmetry and the Ising Model

Let us begin with the concept of a disordered state, as it appears in the Ising model of ferromagnetism. We know that certain materials (e.g., iron) can be magnetized at low temperatures, but above a certain critical temperature, known as the "*Curie Temperature*," the magnetic moment disappears, at least in the absence of an external field. The tendency to retain a magnetic moment at low temperature is due to an interaction between neighboring atoms in the solid, whose potential energy is lowered if the magnetic moments of the neighboring atoms are aligned. The Ising model is a simplified picture of this situation. We imagine that the solid is a cubic array of atoms, and that each atom can be in one of two physical states: "spin up" or "spin down," with the magnetic moment oriented in the direction of the spin. The Hamiltonian of the Ising model is

$$H = -J \sum_x \sum_{\mu=1}^{D} s(x) s(x + \hat{\mu}), \tag{2.1}$$

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where $s(x) = 1$ represents an atom at point x with spin up, $s(x) = -1$ represents spin down, and J is a positive constant.

At low temperatures, in any dimension $D > 1$, the system tends to be in an "*ordered state*," meaning that most spins tend to point in the same direction. The magnetization (average $s(x)$) is non-zero. At high temperatures the system is in a "*disordered state*," in which the average spin is zero, corresponding to a vanishing magnetization. According to the usual principles of statistical mechanics, the probability of any given spin configuration $\{s(x)\}$ at a particular temperature T is

$$\text{Prob}[\{s(x)\}] = \frac{1}{Z}\exp\left[-\frac{H}{kT}\right] \tag{2.2}$$

and

$$Z = \sum_{\{s(x)\}} \exp\left[\beta \sum_x \sum_{\mu=1}^{D} s(x)s(x+\hat{\mu})\right] \tag{2.3}$$

where $\beta \equiv J/kT$. Now observe that the Hamiltonian $H[\{s(x)\}]$, and the probability distribution Prob$[\{s(x)\}]$, are left unchanged by the transformation of each spin by

$$s(x) \to s'(x) = zs(x) \quad \text{where } z = \pm 1. \tag{2.4}$$

Although the transformation with $z = +1$ does nothing to the spins, we include it because the two transformations $\{1, \ -1\}$ together form a group, known as Z_2. The operation (2.4) is called a *"global"* transformation, because every spin $s(x)$ at every location x is transformed in the same way, by the same factor z.

Now it is obvious that the probability distribution is invariant with respect to the global Z_2 transformations, i.e., Prob$[\{s(x)\}]$ = Prob$[\{zs(x)\}]$, and therefore the average spin

$$\langle s \rangle = \sum_{\{s(x)\}} \frac{1}{N_{spins}} \left(\sum_{x'} s(x') \right) \text{Prob}[\{s(x)\}] \tag{2.5}$$

must equal zero. After all, to any given spin configuration $\{s(x)\}$ contributing to the sum, with some average spin s_{av}, there is another configuration with spins $\{-s(x)\}$, with average spin $-s_{av}$, which contributes with exactly the same probability. From this argument, it would appear that magnets are impossible. And in a sense that's true . . . *permanent* magnets, permanent at a finite temperature for infinite time, *are* impossible! Suppose spins are aligned, at some low temperature, with $s_{av} > 0$. There will be small thermal fluctuations, and the s_{av} will vary a little from configuration to configuration, but in general, at low temperatures, s_{av} will be positive for a very long time. A thermal fluctuation which would flip enough spins so that s_{av} becomes negative is very unlikely. Nevertheless, providing the number of spins N_{spins} is finite, and we wait long enough, at some point one of these vastly unlikely fluctuations will occur, and then s_{av} will be negative, again for a very,

very long, yet finite time. Averaged over sufficiently long time scales, the mean magnetization is zero. But the time between such large fluctuations grows exponentially with the number of spins, and for real ferromagnets of macroscopic size the time between flipping the overall magnetic moment, just by thermal fluctuations, would certainly exceed the age of the universe.

So using (2.5) gives a result which is formally correct at low temperatures, yet wrong for all "practical" purposes. It is therefore useful to introduce an external magnetic field h

$$\begin{aligned} H_h &= -J \sum_x \sum_{\mu=1}^{D} s(x)s(x+\hat{\mu}) - h \sum_x s(x) \\ Z_h &= \sum_{\{s(x)\}} \exp[-H_h/kT] \end{aligned} \tag{2.6}$$

so that $\langle s \rangle \neq 0$ at any temperature, and then consider what happens in the pair of limits where we take first the volume (number of spins) infinite, and then reduce h to zero

$$\begin{aligned} m = \lim_{h\to 0} \lim_{N_{spins}\to\infty} \frac{1}{Z_h} \sum_{\{s(x)\}} \left(\frac{1}{N_{spins}} \sum_{x'} s(x') \right) \\ \times \exp\left[\beta \sum_{x,\mu} s(x)s(x+\hat{\mu}) + \frac{h}{kT} \sum_x s(x) \right]. \end{aligned} \tag{2.7}$$

In this pair of limits, done in the order shown, it is possible that $m = \langle s \rangle \neq 0$. When this is the case, we say that the Z_2 global symmetry is "*spontaneously broken*." The term means that despite the invariance of the Hamiltonian, an observable (such as average magnetization) which is not invariant under the symmetry can nevertheless come out with a non-zero expectation value. At high temperatures, even in the limits shown, the magnetization m vanishes. In that case we say that the Z_2 symmetry is unbroken, and the spin system is in the symmetric phase. In general, in the unbroken symmetry phase, the symmetry of the Hamiltonian is reflected in the expectation values: the expectation value vanishes for any quantity which is not invariant with respect to the symmetry group. In the broken phase, non-invariant observables can have non-zero expectation values in the appropriate infinite volume limit.

We also say that the low-temperature, broken phase is an *"ordered"* phase, while the high-temperature, symmetric phase is the "*disordered*" phase. In the case of the Ising model, the term "order" obviously refers to the fact that the spins are not randomly oriented, but point, on average, in one of the two possible directions. This is not true in the disordered phase, although if a given spin in a typical configuration is pointing up, its immediate neighbors are more likely to point up than down. This correlation, however, falls off exponentially with distance between spins. The quantitative measure is provided by the correlation function of the spins at two sites,

loosely denoted "0" and "R," which are a distance R apart in units of the inter-atomic spacing. The correlation function is defined as

$$G(R) = \langle s(0)s(R)\rangle = \frac{1}{Z}\sum_{\{s(x)\}} s(0)s(R)\exp\left[\beta\sum_{x,\mu} s(x)s(x+\hat{\mu})\right]. \tag{2.8}$$

In the disordered phase $G(R) \sim \exp[-R/l]$, where length l is known as the "*correlation length*." In the ordered phase, $G(R) \to m^2$ as $R \to \infty$. It turns out that in $D = 1$ dimension, a spin system is in the disordered phase at *any* finite temperature; there is no phase transition between an ordered and a disordered phase, i.e., no non-zero Curie temperature. The symmetry-breaking transition makes its appearance for any dimension greater than one.

The existence of (at least) two phases, broken and unbroken, ordered and disordered, is characteristic of the statistical mechanics of most many body Hamiltonians that are invariant under some global symmetry.

2.2 Gauge Invariance: the Unbreakable Symmetry

The spin system which was just described is an example of a lattice field theory. The number of dynamical degrees of freedom (spins, in the Ising model) is discrete; each field component interacts with only a few neighbors, and they are usually (but not always) arranged on a square, cubic, or hypercubic lattice, in $D = 2$, 3, 4 dimensions, respectively, as illustrated in Fig. 2.1 for $D = 3$ dimensions. The points of the lattice are known as *sites*, the lines joining neighboring sites are *links*, and, on a hypercubic lattice, the little squares joined by four connecting links are known as *plaquettes*.

A *gauge transformation* is a position-dependent transformation of the degrees of freedom, in which the transformation can be chosen independently at each site.

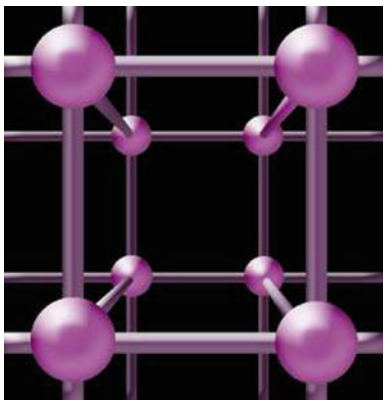


Fig. 2.1 A lattice is a discretization of spacetime. Gauge fields are associated with the links between lattice sites

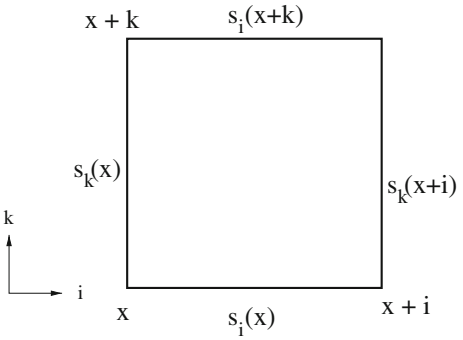


Fig. 2.2 A plaquette is the smallest loop on the lattice. The gauge field action is proportional to a product of link variables around each plaquette, summed over all plaquettes

The trick is to associate the dynamical degrees of freedom with the links of the lattice, while the gauge transformation is specified at each site. A link can be uniquely specified by a lattice site x and direction μ; the link then runs between sites x and $x + \hat{\mu}$. The Hamiltonian for the gauge-invariant Ising model, also known as Z_2 lattice gauge theory, is

$$H = -J \sum_{x} \sum_{\mu=1}^{D-1} \sum_{\nu > \mu}^{D} s_\mu(x) s_\nu(x + \hat{\mu}) s_\mu(x + \hat{\nu}) s_\nu(x). \tag{2.9}$$

The sum is over all plaquettes (Fig. 2.2), and the quantity which is summed is the product of spin variables on links around each plaquette. This Hamiltonian is invariant under the gauge transformation of each link variable $s_\mu(x)$ by

$$s_\mu(x) \rightarrow z(x) s_\mu(x) z(x + \hat{\mu}), \tag{2.10}$$

where the $z(x) = \pm 1$ can be chosen independently at each site; i.e., gauge transformations are *local* transformations, and gauge symmetry is a local symmetry. The local Z_2 gauge symmetry is vastly larger than the global Z_2 symmetry of the Ising model, and this huge expansion of the symmetry can be shown to have the following consequence [1]:

Elitzur's Theorem *A local gauge symmetry cannot break spontaneously. The expectation value of any gauge non-invariant local observable must vanish.*

This means there is no analog of magnetization; e.g., the expectation value of a spin on a link $\langle s_\mu(x) \rangle$ is zero, even if we introduce an external field h (which explicitly breaks the gauge symmetry) and then carefully take first the infinite volume, and then the $h \rightarrow 0$ limits. We must look, instead, to gauge-invariant observables, which are unaffected by gauge transformations. These can be constructed by taking the product of spins on links around a closed loop C, known as a "*Wilson loop*"[1]

[1] The construction was first introduced by Wegner [2].

$$W(C) = \left\langle \prod_{(x,\mu)\in C} s_\mu(x) \right\rangle. \tag{2.11}$$

The product of link variables around a plaquette, found in the gauge-invariant Hamiltonian (2.9), is just a particular example of a Wilson loop.

We can generalize the Z_2 construction to any symmetry group G, just by choosing link variables which are elements of G. Choosing G to be $SU(3)$, or $SU(2) \times U(1)$, we obtain the action of the gauge sector of the strong or electroweak interactions, in a discretized spacetime which allows for numerical simulations. A simple example is $G = U(1)$, and the corresponding gauge theory is known as *compact QED*. In this example the link variables are elements of the $U(1)$ group

$$U_\mu(x) = \exp[iaeA_\mu(x)], \quad A_\mu(x) \in \left[-\frac{\pi}{ae}, \frac{\pi}{ae}\right], \tag{2.12}$$

where a is the lattice spacing, i.e., the distance (in meters, fermis, parsecs, or whatever) between neighboring sites on the lattice, and e is the unit of electric charge. The probability distribution of lattice gauge field configurations is

$$\text{Prob}[\{U_\mu(x)\}] = \frac{1}{Z} e^{-S[U]}, \tag{2.13}$$

where $S[U]$ is known as the "Euclidean action" (to be explained below), and is given by

$$S[U] = -\frac{\beta}{2} \sum_{x,\mu<\nu} U_\mu(x) U_\nu(x+\hat{\mu}) U_\mu^*(x+\hat{\nu}) U_\nu^*(x) + \text{c.c.} \tag{2.14}$$

It is customary to associate the link variable $U_\mu(x)$ with a line running from site x to site $x + \widehat{\mu}$ in the positive μ-direction, and the complex conjugate $U_\mu^*(x)$ (or, for matrix-valued variables, the Hermitian conjugate $U_\mu^\dagger(x)$), with a line running from site $x + \widehat{\mu}$ to x in the negative μ-direction, as shown in Fig. 2.3. In that case the product of link variables shown in (2.14), and its complex conjugate, can be visualized as Wilson loops running along the same plaquette in opposite directions. The Euclidean action $S[U]$ is invariant under local gauge transformations of the form

$$U_\mu(x) \to e^{i\theta(x)} U_\mu(x) e^{-i\theta(x+\hat{\mu})}, \tag{2.15}$$

where the set of transformations $g(x) = \exp[i\theta(x)]$ belongs to the compact $U(1)$ gauge group. To see why this theory is called compact QED, make the Taylor expansion

$$U_\mu(x) = 1 + iaeA_\mu(x) - \frac{1}{2} a^2 e^2 A_\mu^2(x) + \cdots \tag{2.16}$$

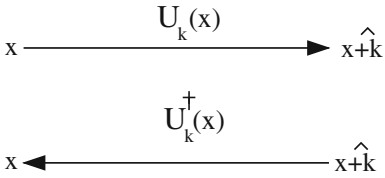


Fig. 2.3 A gauge field variable $U_\mu(x)$ is represented graphically by a directed line joining points x and $x+\hat{\mu}$ in the positive μ-direction. The hermitian conjugate $U_\mu^\dagger(x)$ is represented by a line joining the same points, but in the opposite direction

Keeping terms in the action to lowest order in lattice spacing (and dropping an irrelevant constant)

$$S = \frac{\beta}{2}\sum_x a^4 e^2 \sum_{\mu<\nu}\left[\frac{A_\nu(x+\hat{\mu})-A_\nu(x)}{a} - \frac{A_\mu(x+\hat{\nu})-A_\mu(x)}{a}\right]^2. \tag{2.17}$$

Identifying $\beta = 1/e^2$, we obtain in the limit $a \to 0$, known as the "*continuum limit*,"

$$S = \int \mathrm{d}^4x \frac{1}{4} F_{\mu\nu}F_{\mu\nu}, \tag{2.18}$$

where $F_{\mu\nu}$ is the standard field strength of electromagnetism. However, in the action S the repeated indices are simply summed over, rather than contracted with a Minkowski metric. That is why S is termed a Euclidean action. It is obtained from the usual action of electromagnetism by rotating the time axis in the complex plane $t \to it$ (it is also necessary to rotate the time component of the vector potential, i.e., $A_0 \to -iA_0$). This is called a "*Wick rotation*," and it essentially takes the metric from Minkowski to Euclidean space. In the continuum limit, the gauge transformation (2.15) is equivalent to the familiar form

$$A_\mu(x) \to A_\mu(x) - \frac{1}{e}\partial_\mu\theta(x). \tag{2.19}$$

These constructions are readily generalized to non-abelian groups of interest, $SU(2)$ and $SU(3)$ in particular. For example, $SU(2)$ is the group of 2×2 unitary matrices of determinant one. We let the link variables be elements of this group

$$\begin{aligned} U_\mu(x) &= e^{iagA_\mu(x)} \\ &= b_0 \mathbb{1}_2 + i\mathbf{b}\cdot\boldsymbol{\sigma}, \quad b_0^2 + \mathbf{b}\cdot\mathbf{b} = 1, \end{aligned} \tag{2.20}$$

where $A_\mu(x)$ is a 2×2 Hermitian matrix

$$A_\mu(x) = A_\mu^a(x)\frac{\sigma_a}{2}, \tag{2.21}$$

and the $\{\sigma_a, a = 1, 2, 3\}$ are the three Pauli spin matrices. The Euclidean action

$$S = -\frac{\beta}{2}\sum_{x,\mu<\nu} \mathrm{Tr}[U_\mu(x)U_\nu(x+\hat{\mu})U_\mu^\dagger(x+\hat{\nu})U_\nu^\dagger(x)] \tag{2.22}$$

is invariant under $SU(2)$ gauge transformations

$$U_\mu(x) \to G(x)U_\mu(x)G^\dagger(x+\hat{\mu}), \quad G \in SU(2). \tag{2.23}$$

$SU(2)$ is a bit special, in the sense that the trace of any group element is real. The generalization to an $SU(N)$ gauge theory is simply to take the real part of the trace, i.e.,

$$S = -\frac{\beta}{2N}\sum_{x,\mu<\nu}\Big\{\mathrm{Tr}[U_\mu(x)U_\nu(x+\hat{\mu})U_\mu^\dagger(x+\hat{\nu})U_\nu^\dagger(x)] + \text{c.c.}\Big\}. \tag{2.24}$$

This is known as the *Wilson action*.

Once again, if we expand $U_\mu(x)$ in powers of $A_\mu(x)$, let $\beta = 2N/g^2$, and keep only terms which survive in the $a \to 0$ limit, we obtain

$$S = \int \mathrm{d}^4x \frac{1}{2}\mathrm{Tr}[F_{\mu\nu}F_{\mu\nu}], \tag{2.25}$$

where this time $F_{\mu\nu}(x)$ is the non-abelian field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu], \tag{2.26}$$

and the lattice gauge transformation (2.23) goes over to the continuum gauge transformation

$$A_\mu(x) \to G(x)A_\mu(x)G^\dagger(x) - \frac{i}{g}G(x)\partial_\mu G^\dagger(x), \tag{2.27}$$

with $G(x)$ an element of the gauge group.

2.3 The Monte Carlo Method

I have already mentioned that the lattice action S is a Euclidean action; spacetime indices are contracted with a Kronecker delta, rather than a Minkowski space metric. This Euclidean space formulation has a great advantage: the field theory can be regarded as a statistical (rather than quantum) system, and this lends itself to a powerful method of numerical simulation known as the lattice Monte Carlo

method. Before describing that method, it is important to understand how a simulation in Euclidean space can yield results which are relevant to the corresponding quantum field theory in Minkowski space.

The connection between the Euclidean path integral, and the Minkowski space Hamiltonian operator H, is provided by the following relation: Suppose Q_t is some observable which depends on the fields at a fixed time t. Then

$$\begin{aligned}\langle Q^\dagger_{t_2} Q_{t_1}\rangle &= \frac{1}{Z}\int DA Q^\dagger_{t_2} Q_{t_1} e^{-S} \\ &= \langle \Psi_0 | Q^\dagger e^{-H(t_2-t_1)} Q | \Psi_0\rangle,\end{aligned} \tag{2.28}$$

where the second line is an operator expression, with Ψ_0 the ground state of Hamiltonian H. With the help of this relation, it becomes possible to calculate properties of the Minkowski space theory, e.g., the low-lying spectrum, and the potential energy between static charges, from the lattice theory formulated in Euclidean space.

In a lattice Monte Carlo calculation, the idea is to replace the integral over all configurations $\{U_\mu(x)\}$, weighted by the probability distribution (2.13), by an average over a finite set of sample lattice configurations $\{U^{(n)}, n = 1, 2, \ldots, N_{conf}\}$, i.e.,

$$\begin{aligned}\langle Q\rangle &= \int DUQ[U]\frac{1}{Z}e^{-S[U]} \\ &\approx \frac{1}{N_{conf}}\sum_{n=1}^{N_{conf}} Q[U^{(n)}],\end{aligned} \tag{2.29}$$

where the sample configurations are generated stochastically, with probability weighting $\text{Prob}\big[\{U_\mu(x)\}\big] \sim \exp[-S[U]]$.

There are various ways to generate configurations with the desired probability weighting. The simplest and most general is known as the "*Metropolis algorithm*," and the procedure is the following: Start with any convenient lattice configuration, e.g., all link variables set to the unit element. Sweep through the lattice link by link. At each link choose at random an element of the gauge group, denoted U', as a possible replacement for the existing link variable. Calculate the change ΔS in the action, if link variable $U_\mu(x)$ were replaced by U'. Then

1. If $\Delta S \leq 0$, set $U_\mu(x) = U'$.
2. Otherwise, for $\Delta S > 0$, use a standard random number generator to obtain a pseudo-random number x, in the interval [0, 1] with uniform weighting. If $x < \exp[-\Delta S]$, set $U_\mu(x) = U'$. Otherwise, leave the link variable unchanged, and go on to the next link.

It can be shown that by iterating this procedure, one can generate as many lattice configurations as one likes with the desired probability weighting. Then any observable, such as a Wilson loop, can be averaged over the N_{conf} sample configurations to give an estimate of the expectation value. The statistical error

inherent in this procedure goes down as $1/\sqrt{N_{conf}}$. While the Metropolis algorithm is easy to explain and easy to implement, it is often not the most efficient method in practice. For the more sophisticated algorithms which are used in modern, large-scale lattice Monte Carlo calculations, the reader should consult, e.g., the text by DeGrand and DeTar [3].

We can use the expectation value of Wilson loops, computed in the Euclidean, statistical theory, to determine the interaction energy of two static, color-charged sources in the Minkowski space quantum theory. This interaction energy is known as the *static quark potential*, whether or not the sources are massive quarks. Consider, e.g., adding a massive scalar field to the gauge theory in D-dimensions, with action

$$S_{matter} = -\sum_{x,\mu}\left(\phi^\dagger(x)U_\mu(x)\phi(x+\hat{\mu}) + \text{c.c.}\right) + \sum_x (m^2+2D)\phi^\dagger(x)\phi(x). \quad (2.30)$$

This action is gauge invariant, given that the scalar field transforms as $\phi(x) \to G(x)\phi(x)$. Let Q_t be an operator which creates a gauge invariant particle–anti-particle state at time t

$$Q_t = \phi^\dagger(0,t)U_i(0,t)U_i(\hat{\imath},t)U_i(2\hat{\imath},t)\ldots U_i((R-1)\hat{\imath},t)\phi(R\hat{\imath},t). \quad (2.31)$$

For $m^2 \gg 1$ the one-link term in S_{matter} can be treated as a small perturbation. In this limit it is straightforward to integrate over ϕ in the functional integral, and we find, to leading order in $1/m^2$

$$\langle Q_T^\dagger Q_0\rangle = \text{const.} \times \langle \text{Tr}[UUUU\ldots UUU]_C\rangle, \quad (2.32)$$

where $[UUU\ldots U]_C$ represents an ordered product of link variables (either U or $U^\dagger$), running counterclockwise around a rectangular $R \times T$ contour C, with the convention shown in Fig. 2.3. The trace of this quantity is a gauge-invariant Wilson loop, with expectation value denoted $W(R, T)$. The brackets $\langle\ldots\rangle$ in this case denote the expectation value of the observable in the gauge theory with no matter field, and the constant is $O(m^{-4(T+1)})$.[2] On the other hand, in the operator formalism we also have

$$\begin{aligned}\langle Q_T^\dagger Q_0\rangle &= \frac{\sum_{nm}\langle 0|Q^\dagger|n\rangle\langle n|e^{-HT}|m\rangle\langle m|Q|0\rangle}{\langle 0|e^{-HT}|0\rangle} \\ &= \sum_n |c_n|^2 e^{-\Delta E_n T} \\ &\sim e^{-\Delta E_{min} T} \quad \text{as } T\to\infty,\end{aligned} \quad (2.33)$$

[2] This leading-order (in $1/m^2$) result is obtained by neglecting the one-link term in S_{matter} everywhere except along the time-like links from $t = 0$ to $t = T$, at $x = 0$ and $x = R$. On these links, expand $\exp[\phi^\dagger U_0\phi + \text{h.c.}] \approx 1 + \phi^\dagger U_0\phi + \text{h.c.}$. Integration over the scalar field then yields the result (2.32).

where the sum is over all energy eigenstates of the Hamiltonian containing two static charges at the given locations, and ΔE_n is the difference in energy of the nth excited state and the vacuum. The minimal energy state containing two static charges, with energy difference ΔE_{min}, dominates the expectation value at large T, and this energy difference is also (apart from R-independent self-energy terms) the interaction energy between the static color-charged sources. Therefore, the relationship between the static quark potential $V(R)$ and the expectation value of the rectangular Wilson loop is

$$W(R,T) \sim e^{-V(R)T} \quad (\text{as } T \to \infty). \tag{2.34}$$

2.4 Possible Phases of a Gauge Theory

Elitzur's theorem tells us that gauge symmetries do not break spontaneously, in the sense that any local, gauge non-invariant observable that could serve as an order parameter must have a vanishing expectation value. There is, nonetheless, a qualitative distinction between different phases of a gauge theory, based on the behavior of large Wilson loops. Let us again consider a large rectangular $R \times T$ loop, with $T \gg R$. There are three possibilities:

- **Massive phase**. The potential is of the Yukawa type

$$V(R) = -g^2 \frac{e^{-mR}}{R} + 2V_0, \tag{2.35}$$

where V_0 is a self-energy term, and the Wilson loop falls off at $R > 1/m$ as $\exp(-2V_0T)$. For a generic, non self-intersecting planar loop C, of extension $\gg 1/m$,

$$W(C) \sim \exp[-V_0 P(C)], \tag{2.36}$$

where $P(C)$ is the perimeter of loop C.
- **Massless phases**. The potential has the form [4, 5]

$$V(R) = -\frac{g^2(R)}{R} + 2V_0, \tag{2.37}$$

and the rectangular loop falls off as $\exp[-2V_0T + g^2(R)T/R]$. The effective coupling $g^2(R)$ can go to a constant (Coulomb phase), or, in gauge theories with massless matter fields, it may decrease logarithmically (free electric phase), or increase logarithmically (free magnetic phase), depending on the type of massless particles in the spectrum. For $R \gg V_0^{-1}$, the Wilson loop again has a perimeter-law falloff.
- **Magnetic disorder phase**. The potential is asymptotically linear

$$V(R) = \sigma R + 2V_0. \tag{2.38}$$

Then the Wilson loop falls off asymptotically as $\exp[-\sigma RT - 2V_0 T]$. More generally, for a planar non self-intersecting loop with minimal area $A(C)$ and perimeter $P(C)$,

$$W(C) \sim \exp[-\sigma A(C) - V_0 P(C)], \tag{2.39}$$

and there is an area-law type of falloff.

The term "magnetic disorder" is a terminology inspired by electromagnetism. Consider any lattice (or continuum) gauge theory, based on an abelian gauge group. Let C be a planar loop, whose minimal area $A(C)$ is subdivided into a set $\{C_i\}$ of n smaller loops, each of area A' as shown in Fig. 2.4. If the orientations of the loops are chosen so that neighboring contours run in opposite directions, it is easy to see that

$$U(C) = \prod_{i=1}^{n} U(C_i). \tag{2.40}$$

Now suppose that if the subareas $\{A_i'\}$ are large enough, the expectation value of the product of loops is equal to the product of expectation values, i.e.,

$$\left\langle \prod_{i=1}^{n} U(C_i) \right\rangle = \prod_{i=1}^{n} \langle U(C_i) \rangle. \tag{2.41}$$

In that case,

$$\begin{aligned} W(C) &= \langle U(C) \rangle \\ &= \exp[-\sigma A(C)], \end{aligned} \tag{2.42}$$

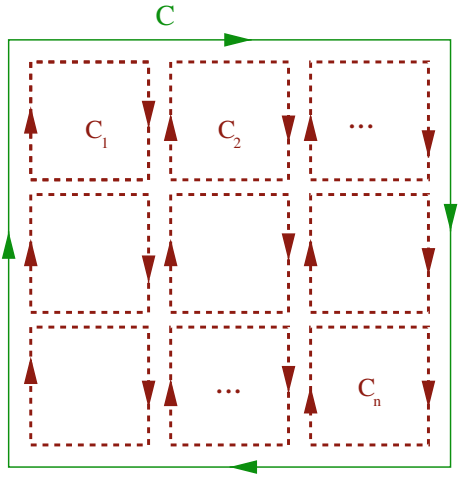


Fig. 2.4 Subdivison of the minimal area of a large loop C into smaller areas bounded by loops $\{C_i\}$. In an abelian gauge theory, the Wilson loop around C is the product of Wilson loops around the $\{C_i\}$

where

$$\sigma = -\frac{\log\langle U(C_i)\rangle}{A'}. \tag{2.43}$$

If we consider varying the size and number of the subloops C_i, it is clear that, so long as the factorization property (2.41) holds, consistency requires that the string tension σ in (2.43) is independent of the subloop area A'.

The relationship (2.40), for the $U(1)$ gauge group, is simply a consequence of Stokes Law, since, in the continuum limit, for a planar loop in, e.g., the ij plane

$$\begin{aligned} U(C) &= \exp\left[ie \oint_C \mathrm{d}x^k A_k(x)\right] \\ &= \exp\left[ie \int \mathrm{d}S_C F_{ij}\right], \end{aligned} \tag{2.44}$$

where the second integration runs over the minimal area of the loop. Then (2.40) follows from the fact that

$$\int \mathrm{d}S_C F_{ij} = \sum_n \int \mathrm{d}S_{C_n} F_{ij}. \tag{2.45}$$

If i, j are spacelike directions, then in $3 + 1$ dimensions the integrals over S_{C_n} just give the magnetic flux passing through the area of the loop C_n.

It is Stoke's Law, in the abelian theories, that tells us that the product of smaller Wilson loops equals the single larger Wilson loop (2.40). The statement that the expectation value of the product of smaller loops is equal to the product of expectation values in (2.41) goes beyond Stoke's Law, and translates into the statement that if the minimal area A' is large enough, then the magnetic fluxes through the loops in the set $\{C_i\}$ are completely uncorrelated. This is the property which leads to an area law falloff for Wilson loops, and which motivates the term "magnetic disorder."

For non-abelian gauge groups, the Stokes law is not so simple (it exists, in fact, in several inequivalent versions, cf. [6–9]), but still boils down to the fact that a large planar Wilson loop can be expressed as the product of Wilson loops over smaller subareas.[3]

In $D = 2$ dimensions it is easy to demonstrate that the only phase that exists is the magnetically disordered phase, for any lattice action of the form (2.24), and for any gauge group. Take the Z_2 gauge theory for simplicity, with

[3] One approach, based on the non-abelian Stokes Law, derives an area law for a large Wilson loop from an assumed finite range behavior of field strength correlators, which means that field strengths are uncorrelated, i.e. disordered, at sufficiently large separations. This "field correlator" approach to magnetic disorder has been pursued by Simonov and co-workers [10].

$$\begin{aligned} S &= -\beta \sum_p s(p) \\ s(p) &= s_1(x) s_2(x+\hat{1}) s_1(x+\hat{2}) s_2(x), \end{aligned} \tag{2.46}$$

where the sum is over all plaquettes p defined on a rectangular lattice of N_p plaquettes, with free, rather than periodic boundary conditions.[4] We use the fact that, for $s(p)$ the product of links around a plaquette p,

$$\exp[\beta s(p)] = \cosh(\beta) + s(p)\sinh(\beta), \tag{2.47}$$

and

$$\sum_{s_\mu = \pm 1} s_\mu = 0, \quad \sum_{s_\mu = \pm 1} 1 = \sum_{s_\mu = \pm 1} s_\mu^2 = 2. \tag{2.48}$$

Then for a loop C enclosing a region S of area $A(C)$

$$W(C) = \frac{\sum_{s_\mu(x)} \left(\prod_{(x',\mu') \in C} s_{\mu'}(x') \right) \prod_p (\cosh(\beta) + s(p)\sinh(\beta))}{\sum_{s_\mu(x)} \prod_p (\cosh(\beta) + s(p)\sinh(\beta))}. \tag{2.49}$$

The only terms which survive in the numerator and denominator have either zero or an even number of link variables at each link, i.e.,

$$\begin{aligned} \text{numerator} &= \sum_{s_\mu(x)} \left(\prod_{(x',\mu') \in C} s_{\mu'}(x') \right) \prod_p (\cosh(\beta) + s(p)\sinh(\beta)) \\ &= \sum_{s_\mu(x)} \left(\prod_{(x',\mu') \in C} s_{\mu'}(x') \right) \prod_{p' \in S} s(p') \sinh(\beta) \prod_{p \notin S} \cosh(\beta) \\ &= \sum_{s_\mu(x)} (\sinh(\beta))^{A(C)} (\cosh(\beta))^{N_p - A(C)} \\ &= 2^{N_p} (\sinh(\beta))^{A(C)} (\cosh(\beta))^{N_p - A(C)}, \\ \text{denominator}(= Z) &= \sum_{s_\mu(x)} \prod_p (\cosh(\beta) + s(p)\sinh(\beta)) \\ &= \sum_{s_\mu(x)} (\cosh(\beta))^{N_p} \\ &= 2^{N_p} (\cosh(\beta))^{N_p}, \end{aligned} \tag{2.50}$$

so that

[4] The Wilson loop calculation is a little easier in two dimensions with free boundary conditions. Periodic boundary conditions introduce a correction which is irrelevant for N_p large.

$$\begin{aligned} W(C) &= (\tanh(\beta))^{A(C)} \\ &= \exp[-\sigma A(C)] \quad \text{with } \sigma = -\log[\tanh(\beta)]. \end{aligned} \tag{2.51}$$

This demonstrates that Wilson loops have an area law falloff, and Z_2 lattice gauge theory is in the magnetically disordered phase, for any value of $\beta > 0$ in $D = 2$ dimensions. The calculation for $U(1)$ and $SU(N)$ gauge groups is similar, with $\cosh(\beta)$ and $\sinh(\beta)$ replaced by other special functions. For abelian gauge groups such as Z_2 and $U(1)$, it is also straightforward to verify the property (2.41); in the case of Z_2 gauge theory both sides of (2.41) equal $\tanh^{A(C)}(\beta)$. For non-abelian groups, an expression of the form (2.41) holds, in $D = 2$ dimensions, for the product of traces $Tr[U(C_i)]$ of loops.

The underlying reason for magnetic disorder in $D = 2$ dimensions is the absence of a Bianchi constraint relating different components of the field strength tensor. In general, the classical gauge field equations in $D = 4$ dimensions are of two sorts. First, there are the genuine equations of motion, which follow from variation of the action with respect to A^a_μ

$$(D^\mu)^{ab} F^b_{\mu\nu} = j^a_\nu, \tag{2.52}$$

where D_μ is the covariant derivative

$$D^{ab}_\mu = \delta^{ab}\partial_\mu - g f^{acb} A^c_\mu, \tag{2.53}$$

and j^a_μ is an external current source. In electromagnetism, where the color indices a, b are absent, and the covariant derivative is just an ordinary derivative, these equations of motion are Gauss's Law and Ampere's Law. But there are also a set of conditions on the field strength tensor which follow as identities from the fact that the field strength is expressed in terms of the 4-vector potential

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu. \tag{2.54}$$

These identities are the Bianchi constraint

$$(D^\mu)^{ab*} F^b_{\mu\nu} = 0, \tag{2.55}$$

where ${}^*F^b_{\mu\nu}$ is the "dual" field-strength tensor

$${}^*F^a_{\mu\nu} = \frac{1}{2}\varepsilon_{\mu\nu\alpha\beta} F^{a\alpha\beta}. \tag{2.56}$$

In electromagnetism, the Bianchi constraint in $D = 4$ dimensions consists of the source-free Maxwell's equations, i.e., Faraday's Law plus the condition $\nabla \cdot B = 0$. In $D = 3$ dimensions there is only the condition $\nabla \cdot B = 0$. In $D = 2$ dimensions the Bianchi constraint is absent.

Many years ago, Halpern [11] showed that it was possible to change integration variables in the functional integral from the gauge potentials $A_\mu(x)$ to the field strength variables $F_{\mu\nu}$, with the result that

$$\begin{aligned} Z &= \int DA_\mu \exp\left[-\int \mathrm{d}^D x \frac{1}{2}\mathrm{Tr}[F_{\mu\nu}^2]\right] \\ &= \int DF_{\mu\nu}\delta[I(F)] \exp\left[-\int \mathrm{d}^D x \frac{1}{2}\mathrm{Tr}[F_{\mu\nu}^2]\right], \end{aligned} \tag{2.57}$$

where $I(F) = 0$ is the Bianchi constraint, expressed entirely in terms of the (abelian or non-abelian) field strengths. In this formulation it is clear that the only thing which correlates the values of the field strength at neighboring points is the Bianchi constraint. In two dimensions this constraint is absent, and field strengths fluctuate independently from one point to another. A similar change of variables, from link to plaquette variables, can be carried out in lattice gauge theory [12], with the similar result that in two dimensions the plaquette variables fluctuate independently. Again using Z_2 lattice gauge theory as an example, we can use the gauge freedom to transform some of the link variables to +1:

$$s_1(x) = 1, \quad s_2(x_1 = 0, x_2) = 1. \tag{2.58}$$

This is a choice of gauge, analogous to the Coulomb or Landau gauge-fixing conditions in electromagnetism. We may then change variables from the remaining link variables $s_2(x)$ to (an equal number of) plaquette variables $s(p) = s_2(x + \hat{1})s_2(x)$, so that

$$Z \to \prod_p \sum_{s(p)=\pm 1} (\cosh(\beta) + s(p)\sinh(\beta)), \tag{2.59}$$

and we see that the probability distribution factorizes into a product of independent distributions for each plaquette. Using the Stokes Law

$$\prod_{(x',\mu')\in C} s_{\mu'}(x') = \prod_{p\in S} s(p) \tag{2.60}$$

we can quickly rederive the result for Wilson loops

$$\begin{aligned} W(C) &= \frac{1}{Z}\prod_{p'\in S}\sum_{s(p')=\pm 1} s(p')(\cosh(\beta) + s(p')\sinh(\beta)) \\ &\quad \times \prod_{p\notin S}\sum_{s(p)=\pm 1} (\cosh(\beta) + s(p)\sinh(\beta)) \\ &= \frac{(2\sinh(\beta))^{A(C)}(2\cosh(\beta))^{N_p - A(C)}}{(2\cosh(\beta))^{N_p}} \\ &= (\tanh(\beta))^{A(C)}. \end{aligned} \tag{2.61}$$

It is clear in this case that the absence of a correlation among even the smallest loops (i.e., plaquette variables) is the crucial ingredient which leads to the area-law expression for Wilson loops of any size. The possibility of changing from link to plaquette variables, without any additional constraint on the plaquette variables, is unique to $D = 2$ dimensions.

In $D > 2$ dimensions, the area law falloff of Wilson loops holds at strong couplings, i.e., $\beta \ll 1$; the calculation, to leading order in β, gives the same result as in $D = 2$ dimensions. At weak couplings, $\beta \gg 1$, the area law certainly does not hold for small loop areas. This is a good thing; we would not like to end up predicting a linear potential between positive and negative electric charges in $3 + 1$ dimensions, and in fact one finds that $U(1)$ gauge theory is in the massless phase in $D = 4$ dimensions at weak couplings. But the area law falloff has been found, via Monte Carlo simulations, to hold for sufficiently large loop areas in non-abelian gauge theories. In the case of the Ising model, the ordered and disordered phases were distinguished by the breaking of a global Z_2 symmetry. Elitzur's theorem assures us that the magnetically disordered phase, and the massive/massless phases, cannot be distinguished by the breaking of the local gauge symmetry. But is there some other global symmetry which would serve this purpose? We will turn to this issue in the next chapter.

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